This equation could also be written as the following, depending on which characteristic you would like to solve for:

```
Current = Voltage ÷ Resistance
```

or

```
Resistance = Voltage ÷ Current
```

Admittedly, this may not be that important to you if all you are doing is hooking up the circuits found in this or other books verbatim, but it could come in handy sometime. Say for example that you buy an LED instead of a lamp to build our basic circuit. The LED itself has very little resistance and might only be rated for +3v at 25 milliamps. If we were to connect the LED straight to the 9v battery, we would have the closest thing to a short circuit and the poor LED wouldn't make it very long. Instead, we need to use a resistor in series with the LED to limit the current flowing through it. Using Ohm's Law, we could solve for this resistance using the following formula:

```
R = 9V ÷ .025A = 360 OHMS
```

By dividing 25 milliamps, or .025 amps, into 9 volts, we find that we need a resistor of about 360 ohms to use the entire available amount of energy. Since a 360-ohm resistor might be a little tricky to find, we can use a 330-ohm resistor instead because that's close enough, and close enough works for us. The LED and resistor in this example are two forms of components that are placed in our circuits. Let's look at others.

Common Components

While there are so many different electronic components, to discuss them all would fill a book larger than this one, so we are going to only look at a few of the very common components here. The idea is to get a sense of what these components look like and generally what kinds of things they do. That way, when you see them discussed or drawn up in a schematic, you'll have an idea of what you're looking at.

Resistors

A resistor limits, or resists, the flow of electricity. They are often useful in a circuit that would not ordinarily place much resistance on the total load, like our LED. In this way, the resistor uses up the current by producing a small amount of heat. Fixed resistors have two wires, have no polarity (so they can be placed in a circuit in any direction), and are rated in ohms, the measurement of resistance, and watts, an indication of the amount of heat the resistor can take. Fixed resistors are marked with either a series of colored bands or other printed markings that determine the resistor's value. Figure 12-3 shows many different kinds of resistors.